Adversarial 2048

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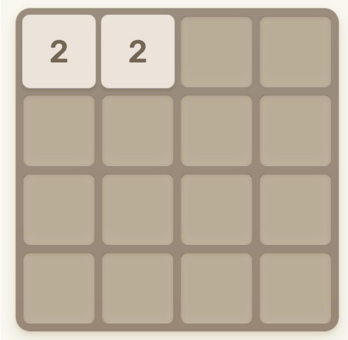
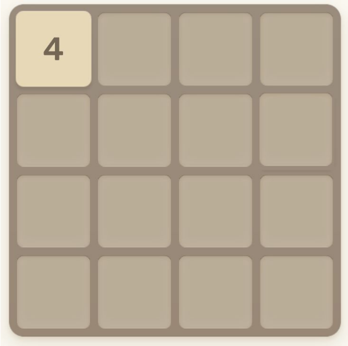
2048

2048 is a simple, single-player web game developed by Gabriele Cirulli in 2014. The goal of the game is to merge numbered squares together with increasing powers of two until you reach 2048 (2^{11}).

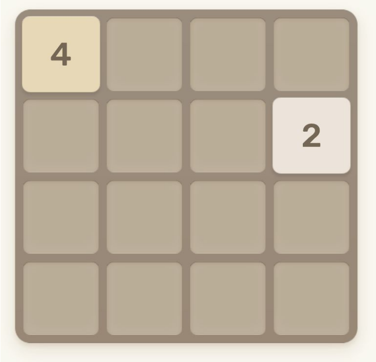
Initial State

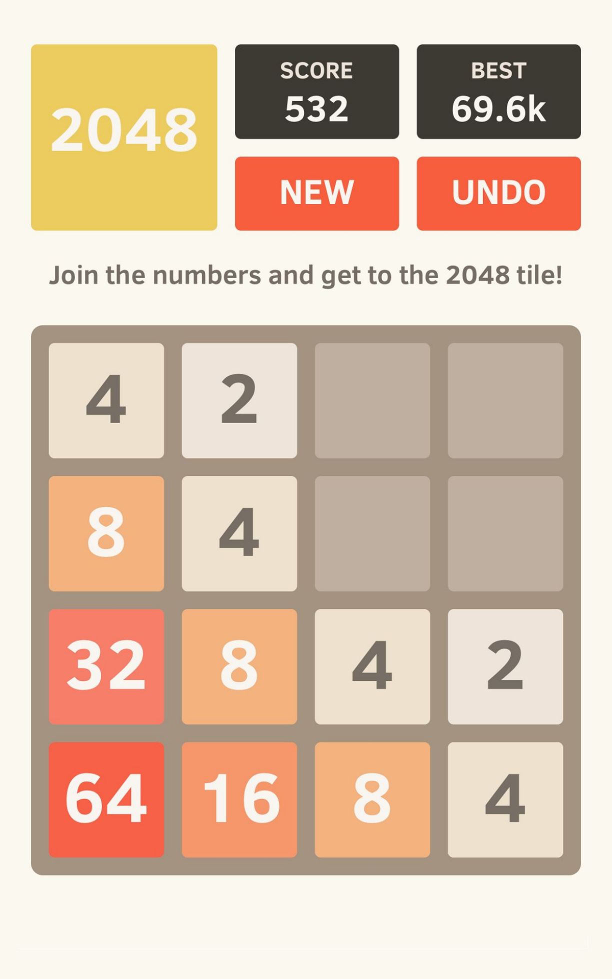


Tilt ←



Tile Generation



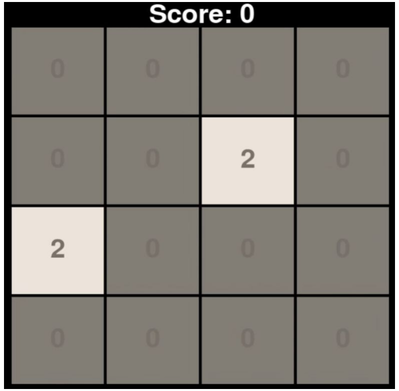


Problem

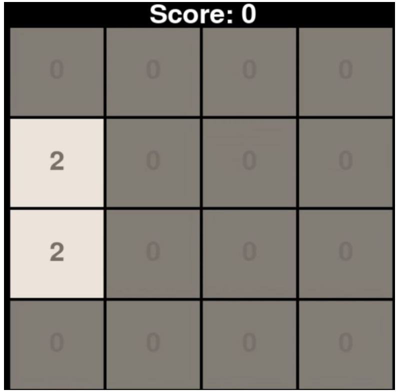
- What would a game of 2048 look like with multiple players?
- Which algorithms would perform the best when playing this hypothetical game?

How the game works

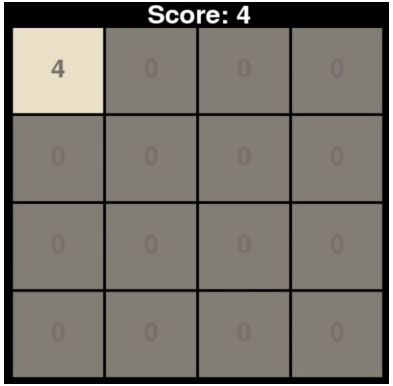
Initial State



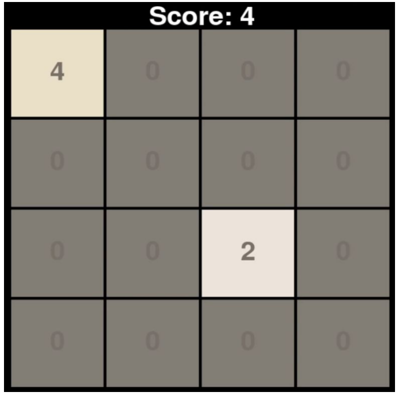
Tilt ←



Tilt ↑

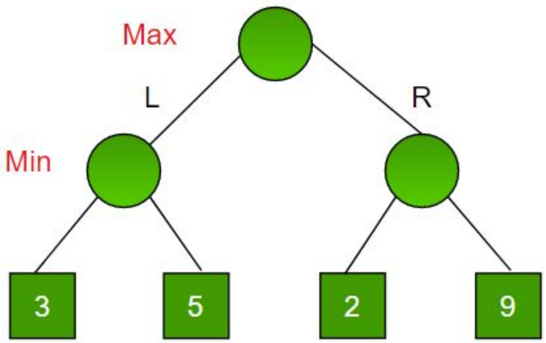


Tile Generation

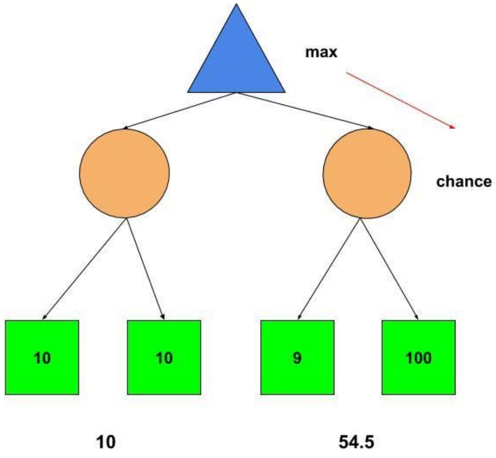


Methods

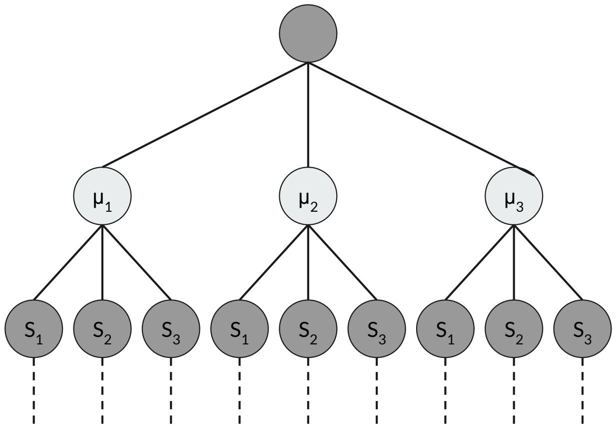
Minimax



Expectimax



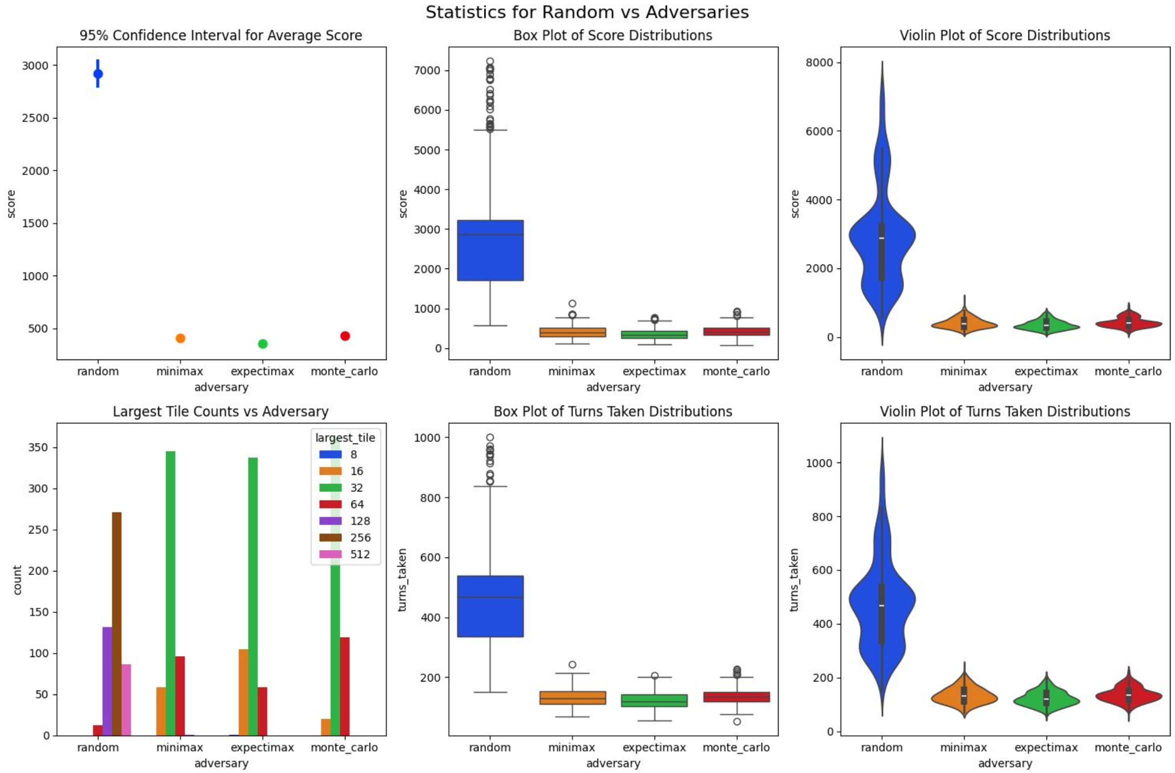
Monte Carlo Simulations



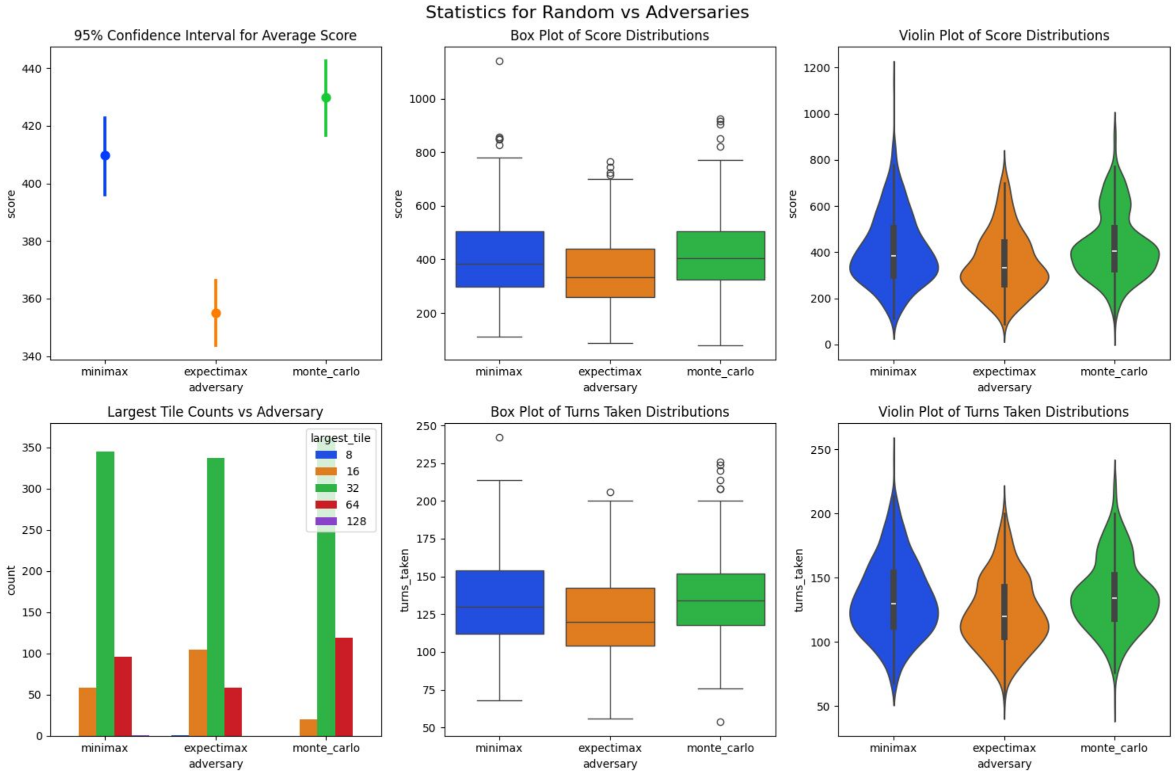
Hypotheses

- When our algorithms play against a random adversary, the score distribution should be right skewed and bimodal.
- Monte-carlo should perform the best in all cases.
- Minimax will perform very poorly, even when compared to a random player.

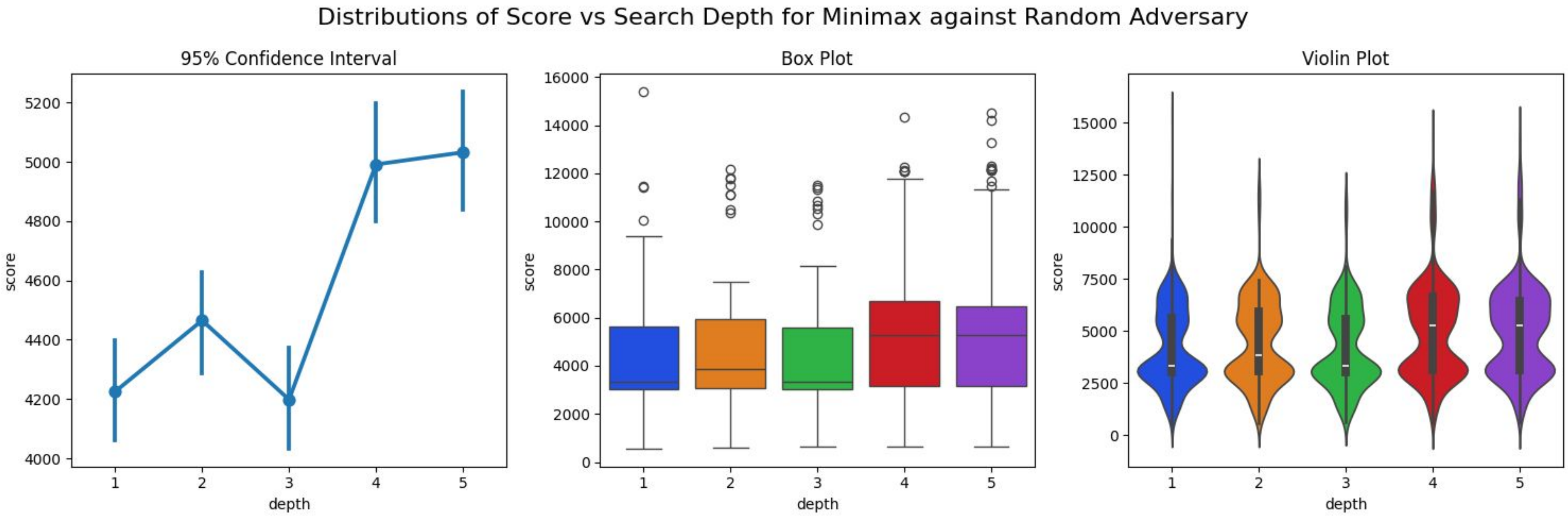
Random Player Effectiveness vs Other Adversaries



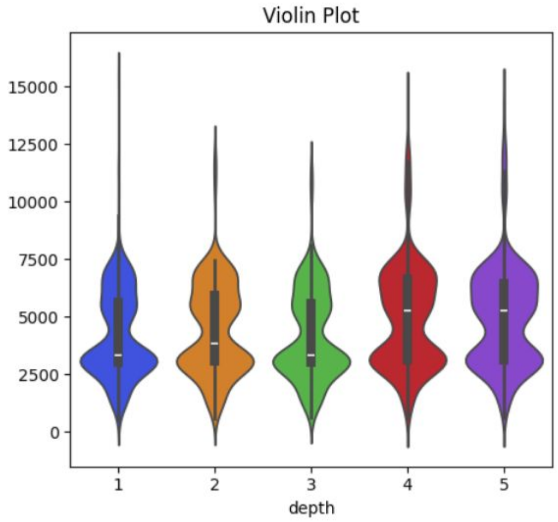
Random Player Effectiveness vs Other Adversaries



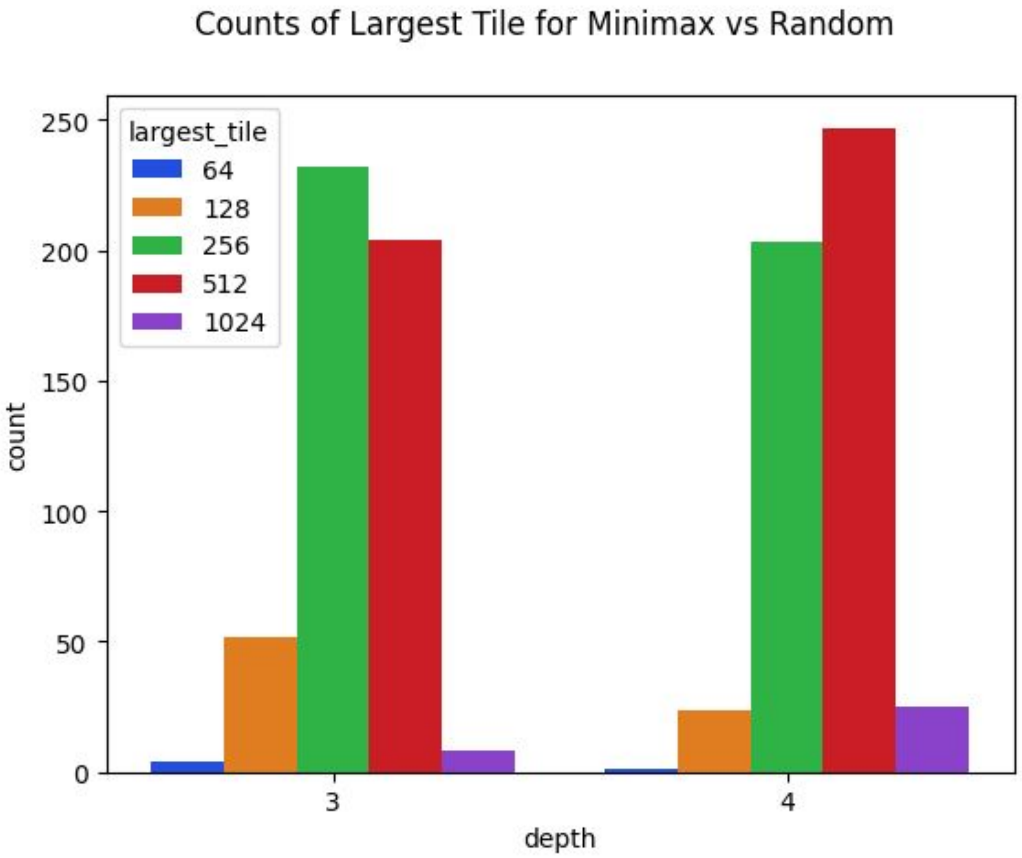
Minimax Effectiveness vs Random Adversary



The “256 wall”

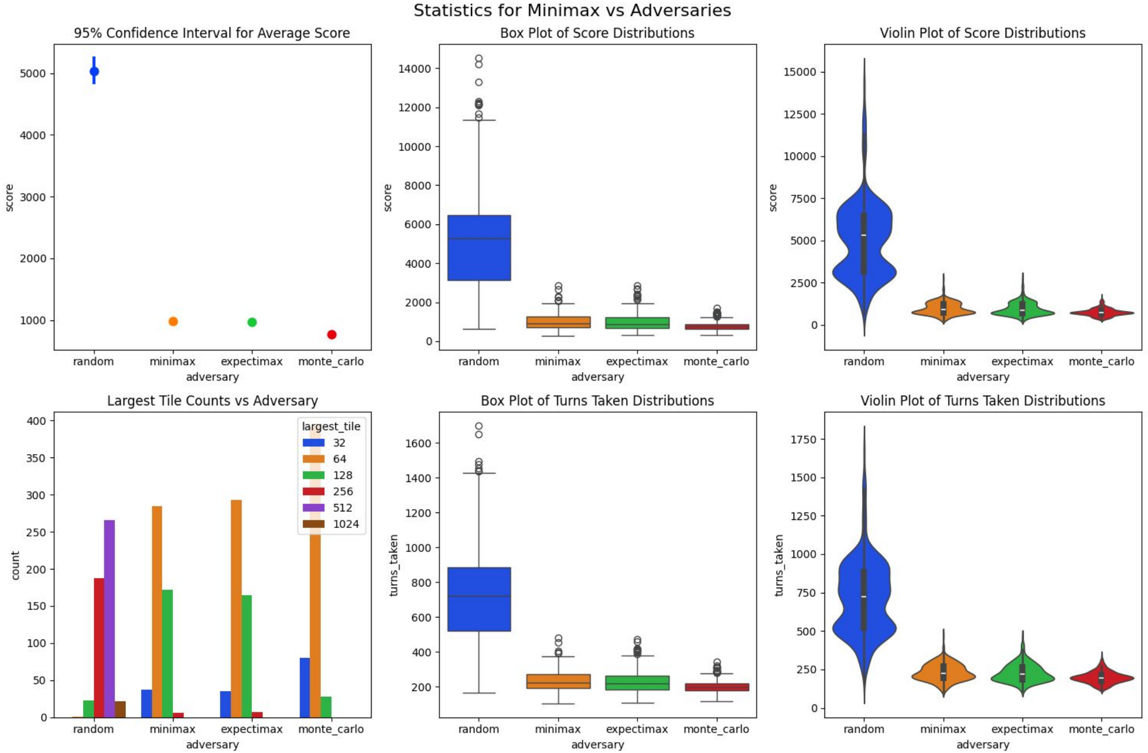


Violin Plot
15000
12500
10000
7500
5000
2500
0
1
2
3
4
5
depth

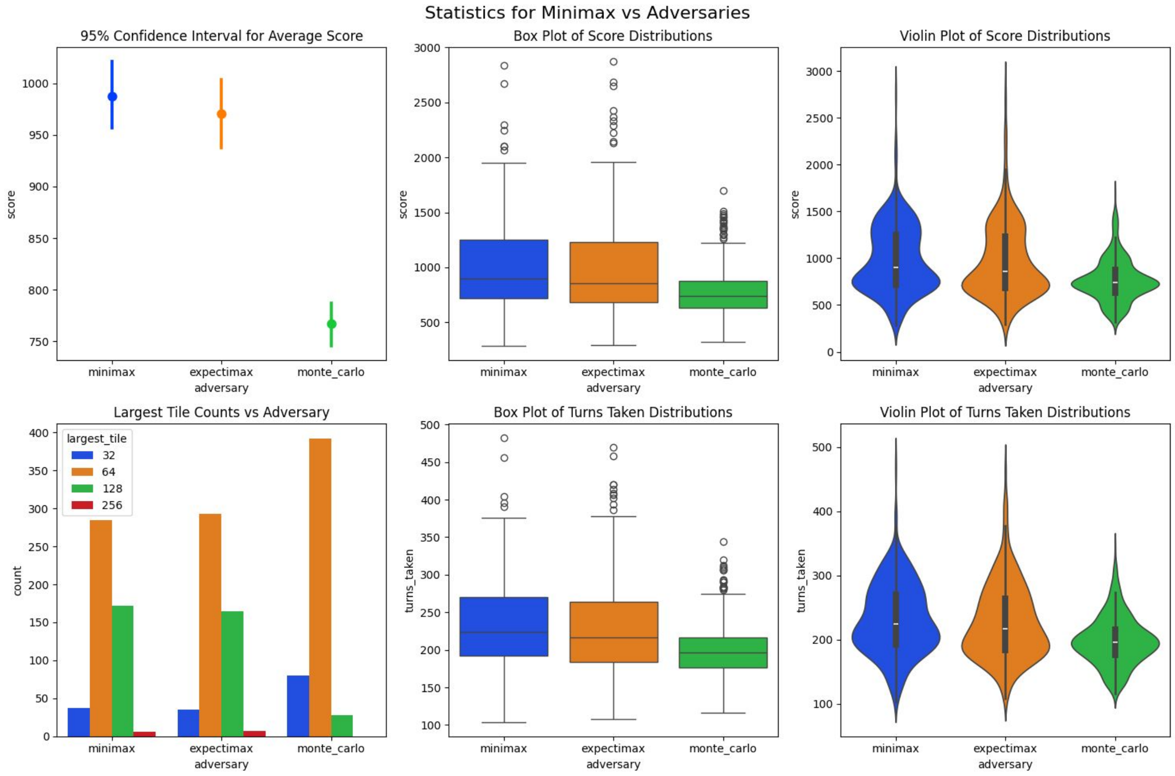


Counts of Largest Tile for Minimax vs Random
largest_tile
64
128
256
512
1024
count
250
200
150
100
50
0
3
4
depth

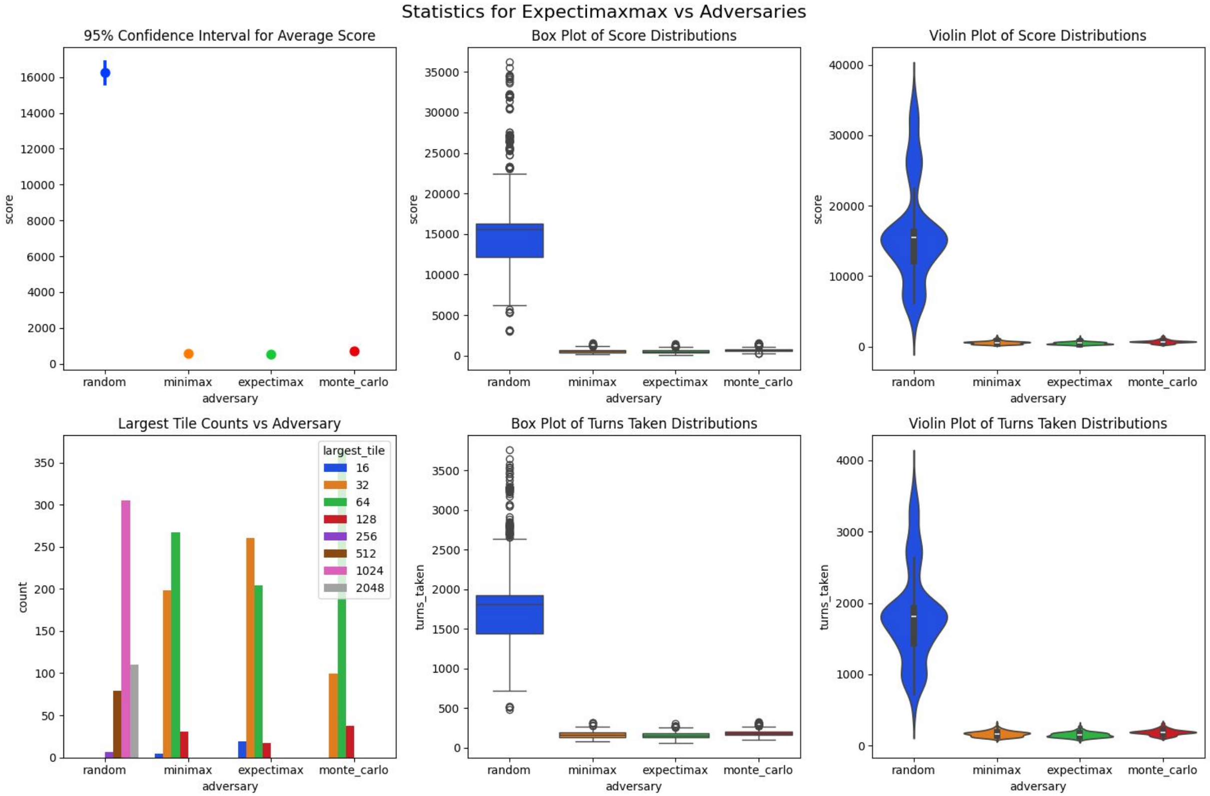
Minimax Effectiveness vs Other Adversaries



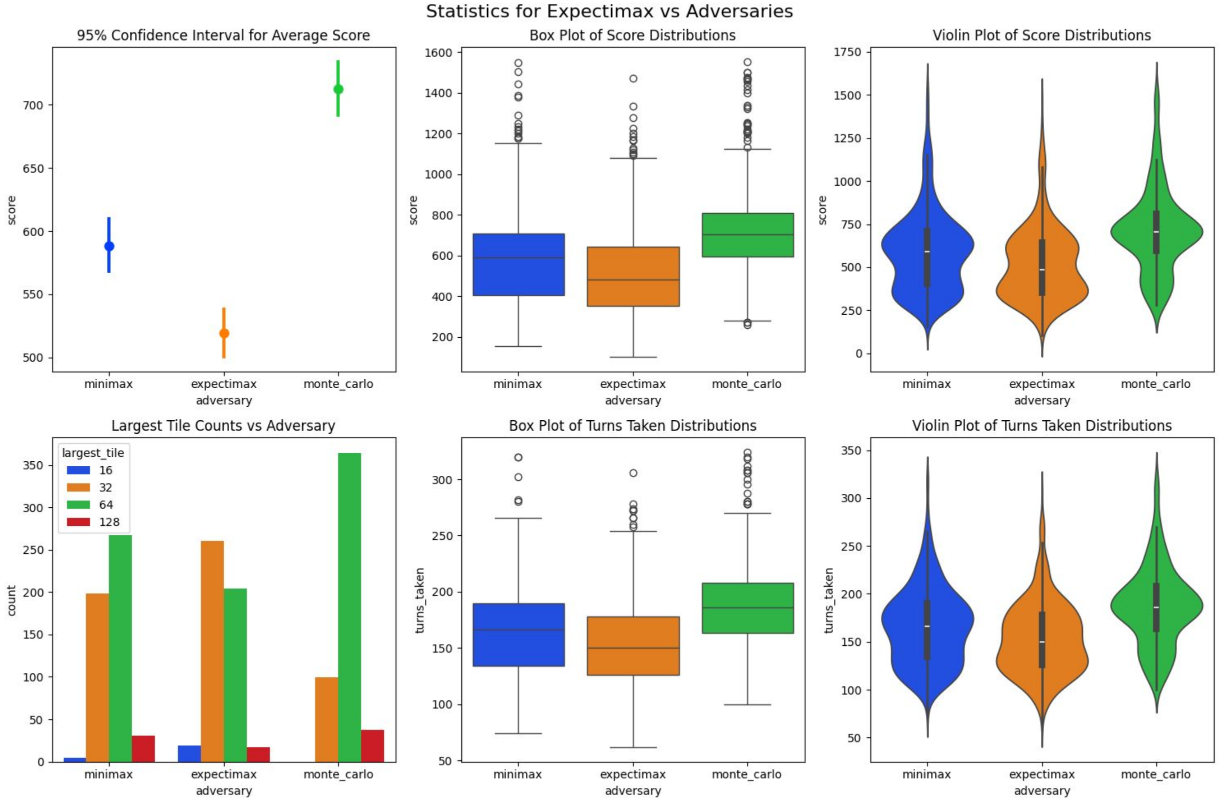
Minimax Effectiveness vs Other Adversaries



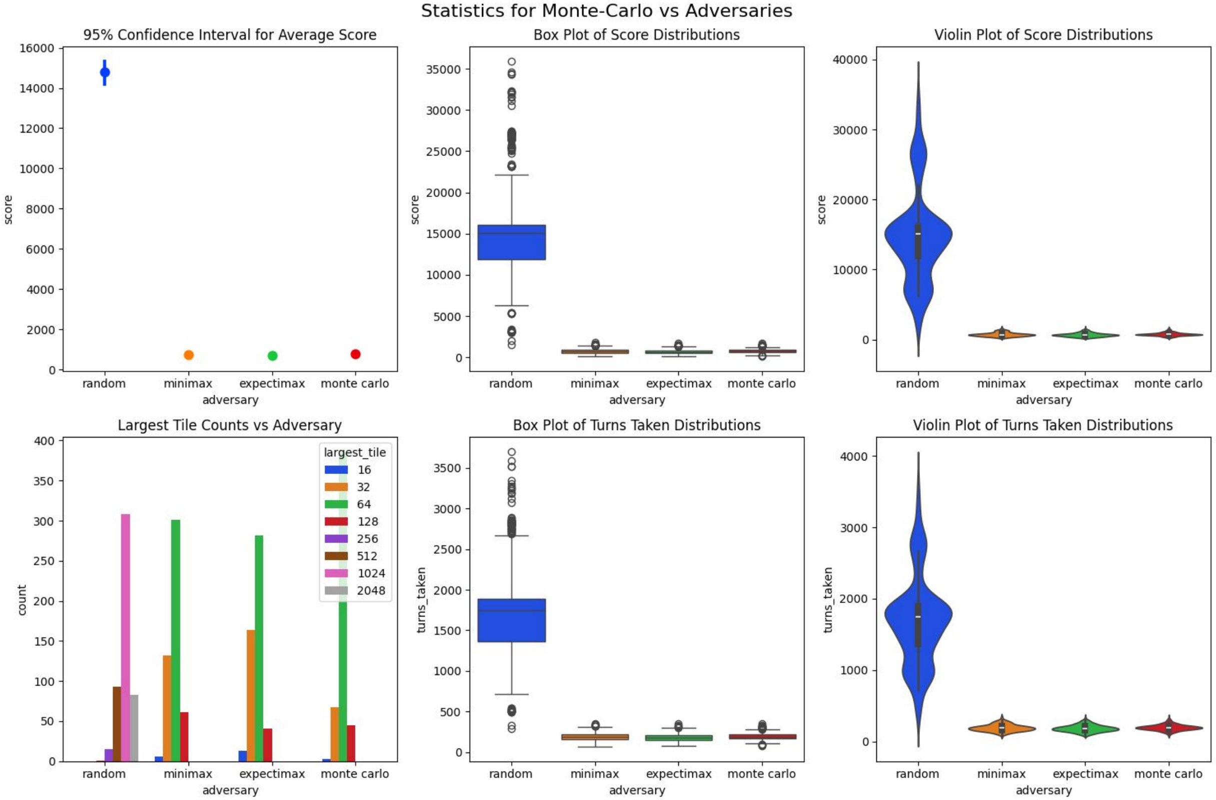
Expectimax Effectiveness vs Other Adversaries



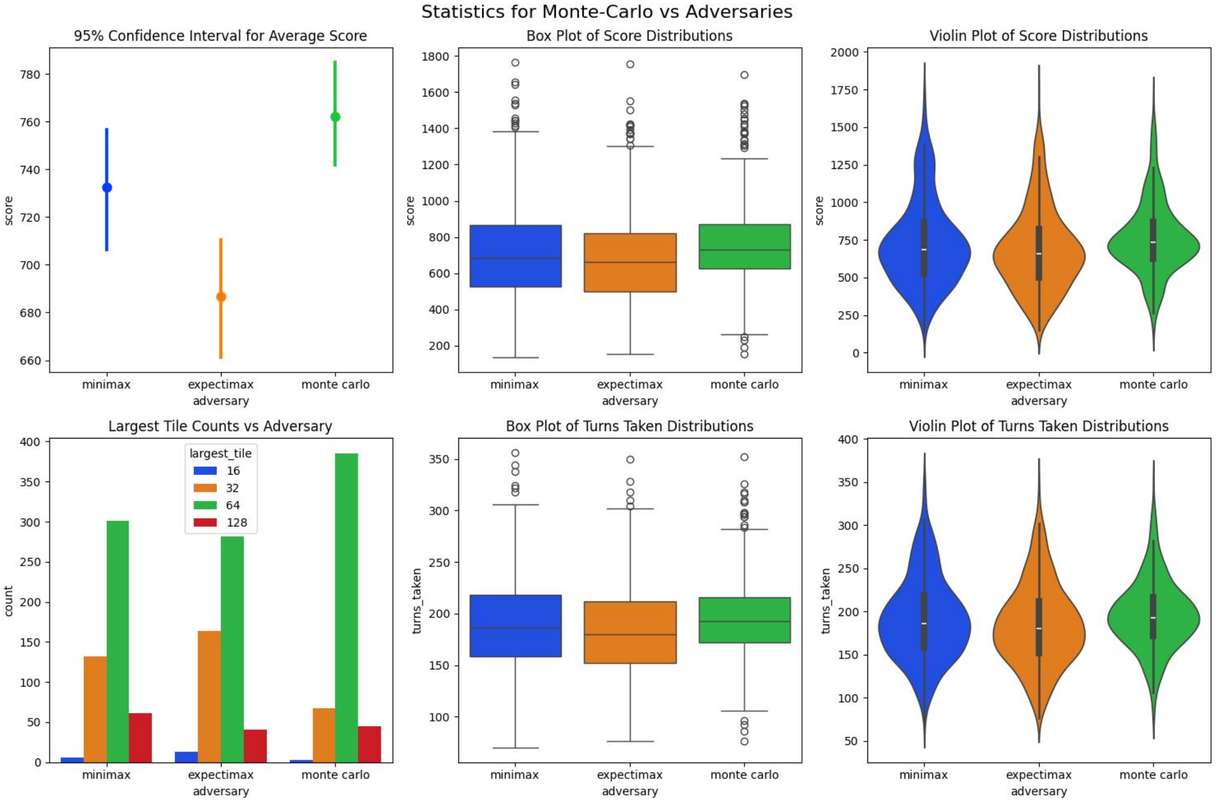
Expectimax Effectiveness vs Other Adversaries



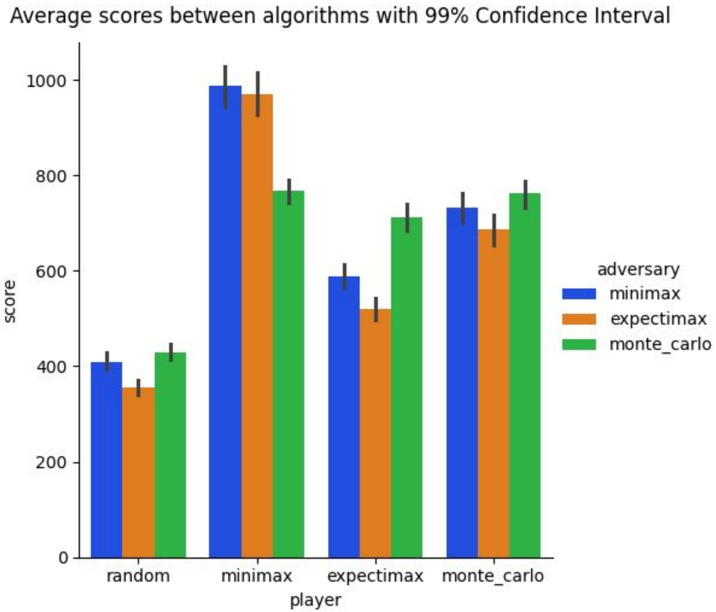
Monte-Carlo Effectiveness vs Other Adversaries

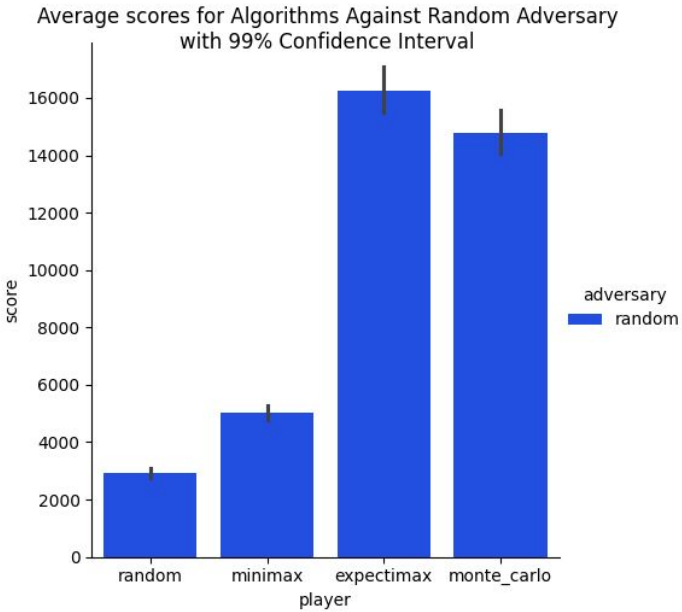


Monte-Carlo Effectiveness vs Other Adversaries



Conclusions





Limitations

- Computationally limited for depth of Expectimax and Minimax, and for rollouts and rollout depth for Monte-Carlo simulations
- Not feasible to run laptops for longer than overnight
- The speed of python for running simulations is somewhat limiting

Demo time!